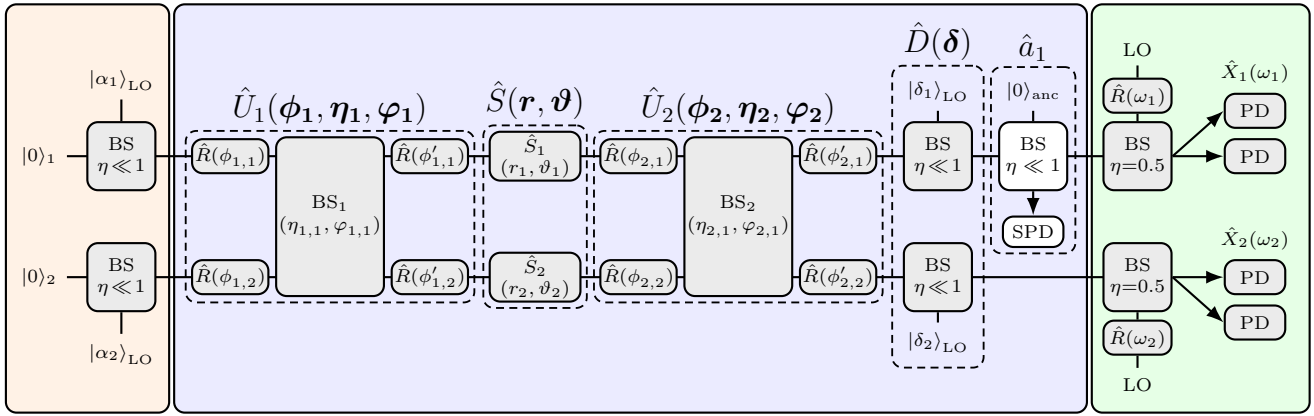

Inputs
$|\boldsymbol{\alpha}\rangle = |\alpha_1, \alpha_2\rangle$
Layer 1
$\hat{L}^{(1)} = \hat{a}_1 \hat{G}(\boldsymbol{\theta_1})$
Outputs
$\langle \hat{x}_j \rangle = \langle \hat{X}_j(0) \rangle$
$|\alpha_1\rangle_{\mathrm{LO}}$
$|0\rangle_1$
BS $\eta \ll 1$
$|0\rangle_2$
BS $\eta \ll 1$
$|\alpha_2\rangle_{\mathrm{LO}}$
$\hat{U}_1(\boldsymbol{\phi_1}, \boldsymbol{\eta_1}, \boldsymbol{\varphi_1})$
$\hat{R}(\phi_{1,1})$
$\hat{R}(\phi_{1,2})$
$\mathrm{BS}_1$ $(\eta_{1,1}, \varphi_{1,1})$
$\hat{R}(\phi'_{1,1})$
$\hat{R}(\phi'_{1,2})$
$\hat{S}(\boldsymbol{r}, \boldsymbol{\vartheta})$
$\hat{S}_1$ $(r_1, \vartheta_1)$
$\hat{S}_2$ $(r_2, \vartheta_2)$
$\hat{U}_2(\boldsymbol{\phi_2}, \boldsymbol{\eta_2}, \boldsymbol{\varphi_2})$
$\hat{R}(\phi_{2,1})$
$\hat{R}(\phi_{2,2})$
$\mathrm{BS}_2$ $(\eta_{2,1}, \varphi_{2,1})$
$\hat{R}(\phi'_{2,1})$
$\hat{R}(\phi'_{2,2})$
$\hat{D}(\boldsymbol{\delta})$
$|\delta_1\rangle_{\mathrm{LO}}$
BS $\eta \ll 1$
$|\delta_2\rangle_{\mathrm{LO}}$
BS $\eta \ll 1$
$\hat{a}_1$
$|0\rangle_{\mathrm{anc}}$
BS $\eta \ll 1$
SPD
LO
$\hat{R}(\omega_1)$
BS $\eta$=0.5
$\hat{X}_1(\omega_1)$
PD
PD
$\hat{X}_2(\omega_2)$
BS $\eta$=0.5
PD
PD
$\hat{R}(\omega_2)$
LO